Lecture 18

Lagrange Multipliers

Constraints: Sometimes you will need to find the maximum or minimum values of functions while the variables have certain conditions or restrictions placed upon them.

For example: Find the maximum / minimum of the function $f(x, y, z) = xy + 2xz + 2yz$, given that,

$xyz - 32 = 0$ $\longrightarrow$ an additional condition placed on x, y, z.

These conditions or restrictions are called constraints.

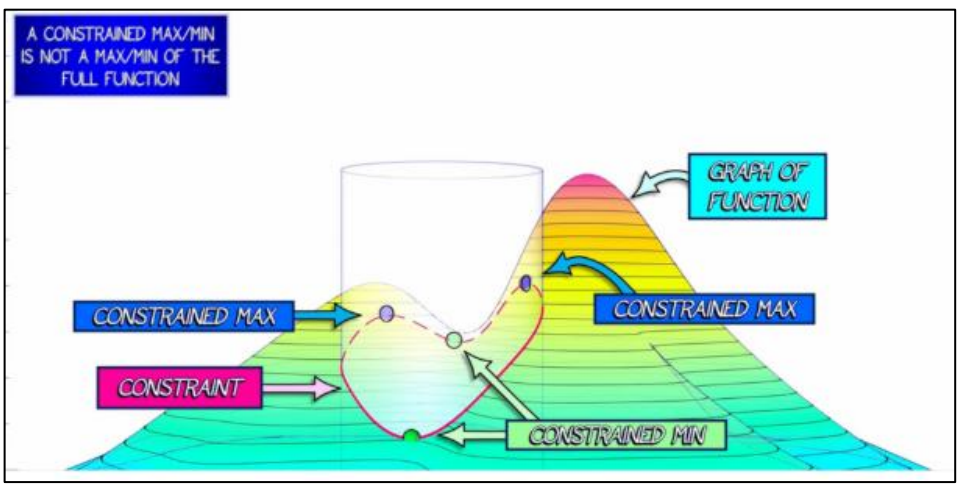
→ Types of problem we will encounter here:

- Two variable extremum problem with one constraint:

Maximize / minimize function $f(x, y)$ or subject to the constraint $g(x, y) = 0$.

- Three variable extremum problem with one constraint:

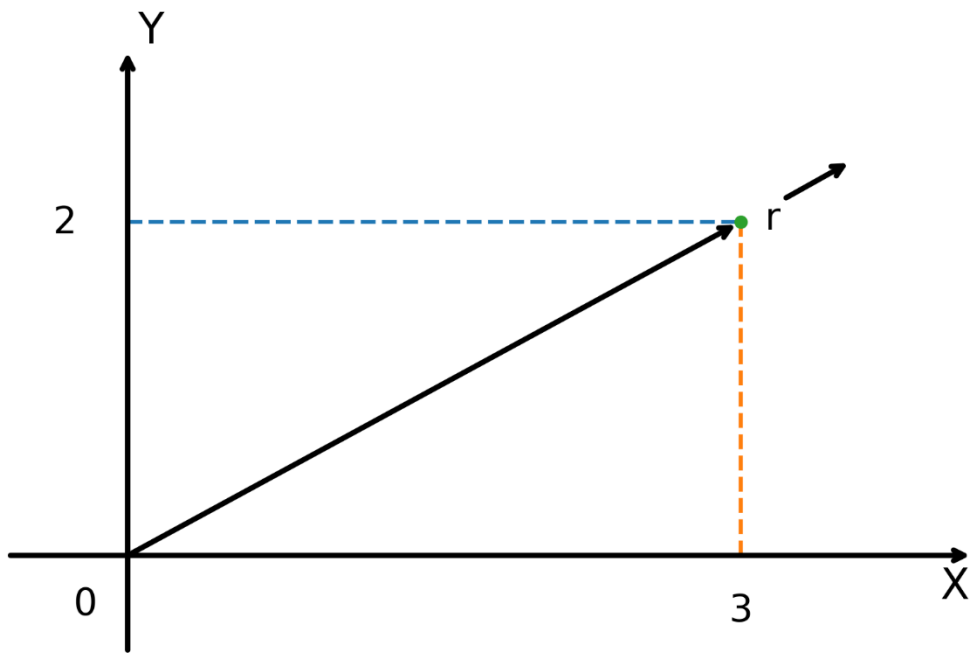
Maximize / minimize function $f(x, y, z)$ subject to the constraint $g(x, y, z) = 0$.



So constrained extrema are points on the constraint curve $g(x, y) = 0$ where the value for $z = f(x, y)$ are the highest or lowest.

Vector: A quantity that has both magnitude and direction.

- In the 2D case, a vector could have two directions: x-direction and the y-direction. They are represented in the form $\langle a, b \rangle$.
  Here 'a' represents how much the vector extended in the x-direction and 'b' represents how much the vector extended in the y-direction.



Here the vector $\vec{r}$ can be represented by $\langle 3,2 \rangle$ or by $3\hat{\imath} + 2\hat{\jmath}$.

- Similarly, in the 3D case, vectors can have three directions: the x-direction, the y-direction and the z-direction. They are represented in the form $\langle a, b, c \rangle$.

Here, 'a' represents how much the vector extended in the x-direction, 'b' represents how much the vector extended in the y-direction and 'c' represents how much the vector extended in the z-direction.

- $\hat{\imath} \rightarrow$ unit vector in the x-direction
- $\hat{\jmath} \rightarrow$ unit vector in the y-direction
- $\hat{k} \rightarrow$ unit vector in the z-direction

<u>Gradient (∇):</u> The gradient of a function $f(x, y)$ at a point (x, y) is given by:

$$\nabla f = \frac{\partial f}{\partial x}\,\hat{\imath} + \frac{\partial f}{\partial y}\,\hat{\jmath}$$

$$\left[\nabla f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle\right]$$

The gradient of a function $f(x, y, z)$ at a point (x, y, z) is given by:

$$\nabla f = \frac{\partial f}{\partial x}\,\hat{\imath} + \frac{\partial f}{\partial y}\,\hat{\jmath} + \frac{\partial f}{\partial z}\,\hat{k}$$

$$\left[\nabla f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle\right]$$

<u>The Lagrange Multiplier:</u> Constrained extrema occur at points on the constraint function where the tangent to the constraint curve is parallel to the tangent of the surface. So, we can say, at optimum,

gradient of $f(x, y) = \text{constant} \times \text{gradient of } g(x, y)$

$\Rightarrow \quad \nabla f(x, y) = \lambda \times \nabla g(x, y)$

Here, λ is a scalar constant and is called the Lagrange multiplier.

Similarly for three variables:

$\nabla f(x, y, z) = \lambda \nabla g(x, y, z)$

Example: Find the maximum and minimum values of $f(x, y, z) = 2x + y - 2z$, subject to the constraint $x^2 + y^2 + z^2 = 4$.

Solution: Given function, $f(x, y, z) = 2x + y - 2z$

Given constraint,

$x^2 + y^2 + z^2 = 4$

$\Rightarrow \; x^2 + y^2 + z^2 - 4 = 0$

Let, $g(x, y, z) = x^2 + y^2 + z^2 - 4$

Now,

$$\nabla f(x, y, z) = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle$$

$$= \langle 2,\ 1,\ -2 \rangle$$

$$\nabla g(x, y, z) = \left\langle \frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right\rangle$$

$$= \langle 2x,\ 2y,\ 2z \rangle$$

Using the method of Lagrange Multipliers:

$\nabla f(x, y, z) = \lambda \nabla g(x, y, z)$

$\Rightarrow \quad \langle 2, 1, -2 \rangle = \lambda \langle 2x, 2y, 2z \rangle = \langle 2\lambda x, 2\lambda y, 2\lambda z \rangle$

Equating the vector components:

$2 = 2\lambda x$ $\Rightarrow \quad \lambda = \frac{1}{x}$	$1 = 2\lambda y$ $\Rightarrow \quad \lambda = \frac{1}{2y}$	$-2 = 2\lambda z$ $\Rightarrow \; \lambda = -\frac{1}{z}$

Combining, we get:

$$\frac{1}{x} = \frac{1}{2y} = -\frac{1}{z} = \lambda$$

Now,

$$\frac{1}{x} = \frac{1}{2y}$$

$$\Rightarrow \ x = 2y$$

$$\Rightarrow \ \frac{x}{2} = y$$

$$\therefore \ y = \frac{x}{2}$$

and

$$\frac{1}{x} = -\frac{1}{z}$$

$$\therefore \ z = -x$$

Substituting the value of y and z in the constraint:

$$x^2 + y^2 + z^2 = 4$$

$$\Rightarrow \ x^2 + \left(\frac{x}{2}\right)^2 + (-x)^2 = 4$$

$$\Rightarrow \ x^2 + \frac{x^2}{4} + x^2 = 4$$

$$\Rightarrow \ \frac{9}{4}x^2 = 4$$

$$\Rightarrow \ x^2 = \frac{16}{9}$$

$$\therefore \ x = \pm\frac{4}{3}$$

If $x = \frac{4}{3}$,

$$y = \frac{x}{2} = \frac{\frac{4}{3}}{2} = \frac{2}{3}$$

and $z = -x = -\frac{4}{3}$

At the point $\left(\frac{4}{3}, \frac{2}{3}, -\frac{4}{3}\right)$, the value of the function f is:

$$f\left(\frac{4}{3}, \frac{2}{3}, -\frac{4}{3}\right)$$

$$= 2\left(\frac{4}{3}\right) + \frac{2}{3} - 2\left(-\frac{4}{3}\right)$$

$= \frac{8}{3} + \frac{2}{3} + \frac{8}{3}$

$= \frac{18}{3}$

$= 6$

If $x = -\frac{4}{3}$,

$y = \frac{x}{2} = \frac{-\frac{4}{3}}{2} = -\frac{2}{3}$

and $z = -x = -\left(-\frac{4}{3}\right) = \frac{4}{3}$

At the point $\left(-\frac{4}{3}, -\frac{2}{3}, \frac{4}{3}\right)$, the value of the function f is:

$f\left(-\frac{4}{3}, -\frac{2}{3}, \frac{4}{3}\right)$

$= 2\left(-\frac{4}{3}\right) + \left(-\frac{2}{3}\right) - 2\left(\frac{4}{3}\right)$

$= -\frac{8}{3} - \frac{2}{3} - \frac{8}{3}$

$= -\frac{18}{3}$

$= -6$

So, $f(x, y, z)$ has maximum value 6 at $\left(\frac{4}{3}, \frac{2}{3}, -\frac{4}{3}\right)$.

and $f(x, y, z)$ has minimum value -6 at $\left(-\frac{4}{3}, -\frac{2}{3}, \frac{4}{3}\right)$.

Example: Use Lagrange multipliers to find the maximum and minimum values of $f(x, y) = 4x^3 + y^2$ subject to the given constraint $2x^2 + y^2 = 1$. Also find the points at which these extreme values occur.

Solution: Given function, $f(x, y) = 4x^3 + y^2$

Given constraint,

$2x^2 + y^2 = 1$

$\Rightarrow \quad 2x^2 + y^2 - 1 = 0$

Let, $g(x, y) = 2x^2 + y^2 - 1$

Now,

$$\nabla f(x,y) = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle$$

$$= \langle 12x^2,\ 2y \rangle$$

And

$$\nabla g(x,y) = \left\langle \frac{\partial g}{\partial x}, \frac{\partial g}{\partial y} \right\rangle$$

$$= \langle 4x,\ 2y \rangle$$

Using the method of Lagrange multipliers:

$\nabla f(x,y) = \lambda\, \nabla g(x,y)$

$\Rightarrow \quad \langle 12x^2,\ 2y \rangle = \lambda \langle 4x,\ 2y \rangle = = \langle 4\lambda x,\ 2\lambda y \rangle$

Equating the vector components:

$12x^2 = 4\lambda x$

$2y = 2\lambda y$

$\Rightarrow \ 2y - 2\lambda y = 0$

$\Rightarrow \ 2y(1-\lambda) = 0$

So, $2y = 0$

$\Rightarrow y = 0$

Or, $1 - \lambda = 0$

$\Rightarrow \lambda = 1$

$\therefore \ 12x^2 = 4x$

$\Rightarrow \ 12x^2 - 4x = 0$

$\Rightarrow \ 4x(3x-1) = 0$

$\Rightarrow \ x(3x-1) = 0$

So, $x = 0$

Or, $3x - 1 = 0$

$\Rightarrow x = \frac{1}{3}$

Substituting the values of x and y in the constraint,

If $x = 0$,

$2x^2 + y^2 = 1$

$\Rightarrow\ 2(0)^2 + y^2 = 1$

$\Rightarrow\ y^2 = 1$

$\Rightarrow\ y = \pm 1$

$\therefore\ (0,1),\ (0,-1)$

If $x = \frac{1}{3}$,

$2x^2 + y^2 = 1$

$\Rightarrow\ 2\left(\frac{1}{3}\right)^2 + y^2 = 1$

$\Rightarrow\ \frac{2}{9} + y^2 = 1$

$\Rightarrow\ y^2 = 1 - \frac{2}{9}$

$\Rightarrow\ y^2 = \frac{7}{9}$

$\Rightarrow\ y = \pm\frac{\sqrt{7}}{3}$

$\therefore\ \left(\frac{1}{3}, \frac{\sqrt{7}}{3}\right),\ \left(\frac{1}{3}, -\frac{\sqrt{7}}{3}\right)$

If $y = 0$,

$2x^2 + y^2 = 1$

$\Rightarrow\ 2x^2 + 0^2 = 1$

$\Rightarrow\ 2x^2 = 1$

$\Rightarrow\ x^2 = \frac{1}{2}$

$\Rightarrow\ x = \pm\frac{1}{\sqrt{2}}$

$\therefore\ \left(\frac{1}{\sqrt{2}}, 0\right),\ \left(-\frac{1}{\sqrt{2}}, 0\right)$

Values of f at these points:

(x, y)	$(0, -1)$	$(0,1)$	$\left(\frac{1}{3}, \frac{\sqrt{7}}{3}\right)$	$\left(\frac{1}{3}, -\frac{\sqrt{7}}{3}\right)$	$\left(\frac{1}{\sqrt{2}}, 0\right)$	$\left(-\frac{1}{\sqrt{2}}, 0\right)$
$f(x, y)$	$4(0)^3 + (-1)^2$ $= 1$	$4(0)^3 + (1)^2$ $= 1$	$4\left(\frac{1}{3}\right)^3$ $+ \left(\frac{\sqrt{7}}{3}\right)^2$ $= \frac{4}{27} + \frac{7}{9}$ $= \frac{25}{27}$ ≈ 0.93	$4\left(\frac{1}{3}\right)^3$ $+ \left(-\frac{\sqrt{7}}{3}\right)^2$ $= \frac{25}{27}$ ≈ 0.93	$4\left(\frac{1}{\sqrt{2}}\right)^3$ $+ 0^2 = \frac{4}{2\sqrt{2}}$ $= \frac{2}{\sqrt{2}} = \sqrt{2}$	$4\left(-\frac{1}{\sqrt{2}}\right)^3$ $+ 0^2$ $= -\frac{4}{2\sqrt{2}}$ $= -\sqrt{2}$

$\therefore$ $f(x, y)$ has maximum value $\sqrt{2}$ at $\left(\frac{1}{\sqrt{2}}, 0\right)$.

$f(x, y)$ has minimum value $-\sqrt{2}$ at $\left(-\frac{1}{\sqrt{2}}, 0\right)$.

H.W. Exercise 13.9 (5-12)